

Lab 3

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```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.1
## Warning: package 'ggplot2' was built under R version 4.5.2
## Warning: package 'tidyr' was built under R version 4.5.1
## Warning: package 'readr' was built under R version 4.5.1
## Warning: package 'purrr' was built under R version 4.5.1
## Warning: package 'dplyr' was built under R version 4.5.1
## Warning: package 'stringr' was built under R version 4.5.1
## Warning: package 'forcats' was built under R version 4.5.1
## Warning: package 'lubridate' was built under R version 4.5.1

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    4.0.2      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(plotrix)
```

```
## Warning: package 'plotrix' was built under R version 4.5.2
```

```
density <- read_csv("../data_raw/Lab3_Density.csv")
```

```
## Rows: 20 Columns: 7
## -- Column specification -----
## Delimiter: ","
## dbf (7): quadrat_number, 50x50cm, 1x1m, 2.5x2.5m, 5x5m, 10x10m, distance_nea...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
time_density <- read_csv("../data_raw/Lab3_densitytime.csv")
```

```
## Rows: 4 Columns: 7
## -- Column specification -----
## Delimiter: ","
```

```
## dbl (7): quadrat_number, 50x50cm, 1x1m, 2.5x2.5m, 5x5m, 10x10m, distance_nea...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
time_frequency <- read_csv("../data_raw/Lab3_frequency_time.csv")

## Rows: 4 Columns: 7
## -- Column specification -----
## Delimiter: ","
## dbl (7): quadrat_number, 50x50cm, 1x1m, 2.5x2.5m, 5x5m, 10x10m, distance_nea...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

1.) Calculate the mean and standard error of mesquite density for each plot size and make a graph of the comparison. (Be sure to convert density)

```
#longer data tables
density_data <- density %>%
  pivot_longer(
    cols = c(`50x50cm`, `1x1m`, `2.5x2.5m`, `5x5m`, `10x10m`),
    names_to = "plot_size",
    values_to = "count"
  ) %>%
  mutate(
    # Set the area in m^2 for each plot size
    area_m2 = case_when(
      plot_size == "50x50cm" ~ 0.25,
      plot_size == "1x1m" ~ 1.00,
      plot_size == "2.5x2.5m" ~ 6.25,
      plot_size == "5x5m" ~ 25.00,
      plot_size == "10x10m" ~ 100.00),
    #convert density
    density = count/area_m2,
    #set as factor to order smallest to largest quadrat and order x-axis titles
    plot_size = factor(plot_size,
      levels = c("50x50cm", "1x1m", "2.5x2.5m", "5x5m", "10x10m")
    )
  )

#standard error and standard deviation
summary_density <- density_data %>%
  group_by(plot_size) %>%
  summarise(
    mean_density = mean(density),
    se_density = sd(density) / sqrt(n())
  )

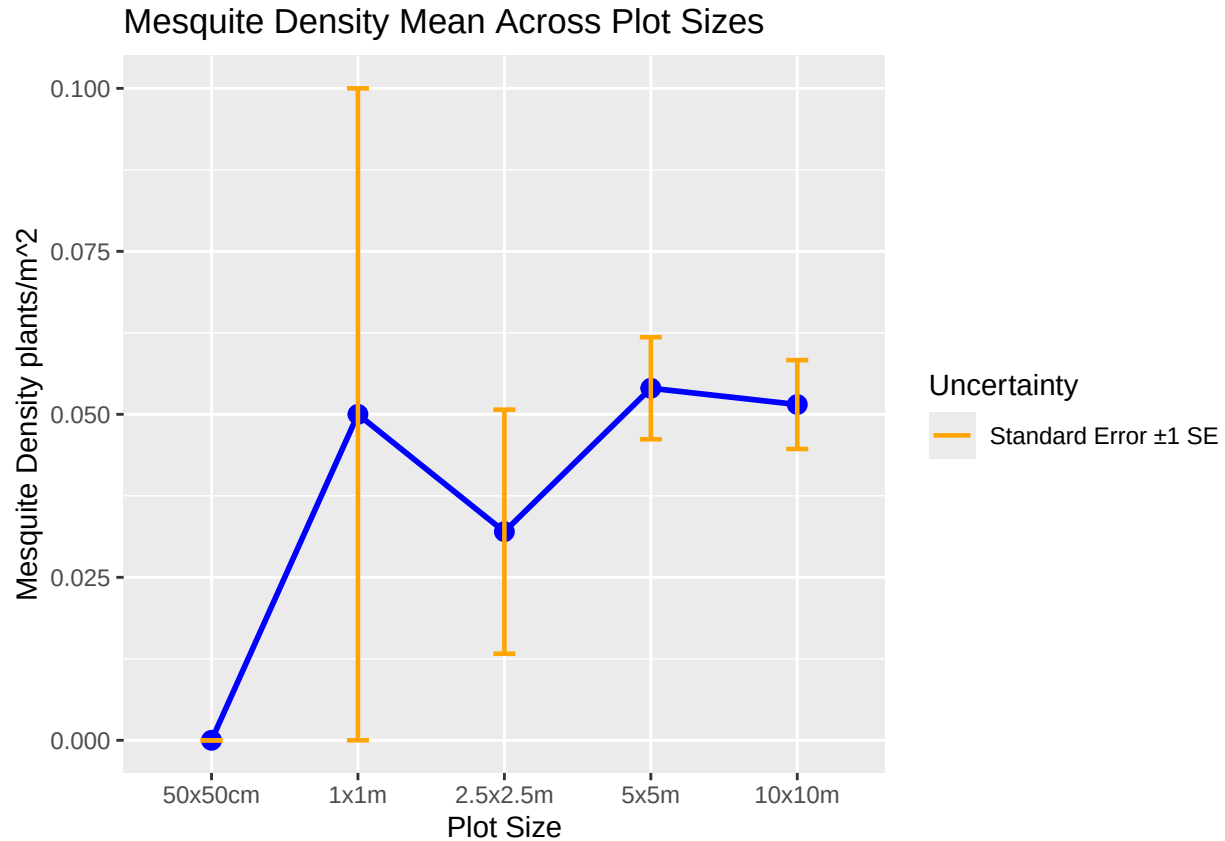
print(summary_density)

## # A tibble: 5 x 3
```

```
##   plot_size mean_density se_density
##   <fct>      <dbl>      <dbl>
## 1 50x50cm      0          0
## 2 1x1m         0.05       0.05
## 3 2.5x2.5m     0.032     0.0187
## 4 5x5m         0.054     0.00783
## 5 10x10m       0.0515    0.00682
```

```
#plot summary stats
ggplot(summary_density, aes(x = plot_size, y = mean_density, group = 1)) +
  geom_line(color = "blue", size = 1) +
  geom_point(color = "blue", size = 3) +
  geom_errorbar(
    aes(ymin = mean_density - se_density, ymax = mean_density + se_density,
        color = "Standard Error ±1 SE"),
    width = 0.15, linewidth = 0.8
  ) +
  scale_color_manual(name = "Uncertainty", values = c("Standard Error ±1 SE" = "orange")) +
  labs(
    title = "Mesquite Density Mean Across Plot Sizes",
    x = "Plot Size",
    y = "Mesquite Density plants/m^2")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



2.) Calculate the average time to sample each plot size.

```
time_density_longer <- time_density %>%
  pivot_longer(
    cols = c(`50x50cm`, `1x1m`, `2.5x2.5m`, `5x5m`, `10x10m`),
    names_to = "plot_size",
    values_to = "count")

time_density_longer <- time_density_longer %>%
  mutate(
    plot_size = factor(
      plot_size,
      levels = c("50x50cm", "1x1m", "2.5x2.5m", "5x5m", "10x10m") ) )

summary_time_density <- time_density_longer %>%
  group_by(plot_size) %>%
  summarise(mean_time = mean(count))

time_frequency_longer <- time_frequency %>%
  pivot_longer(
    cols = c(`50x50cm`, `1x1m`, `2.5x2.5m`, `5x5m`, `10x10m`),
    names_to = "plot_size",
    values_to = c("count"))

time_frequency_longer <- time_frequency_longer %>%
```

```

mutate(
  plot_size = factor( plot_size,
    levels = c("50x50cm", "1x1m", "2.5x2.5m", "5x5m", "10x10m")
  ) )

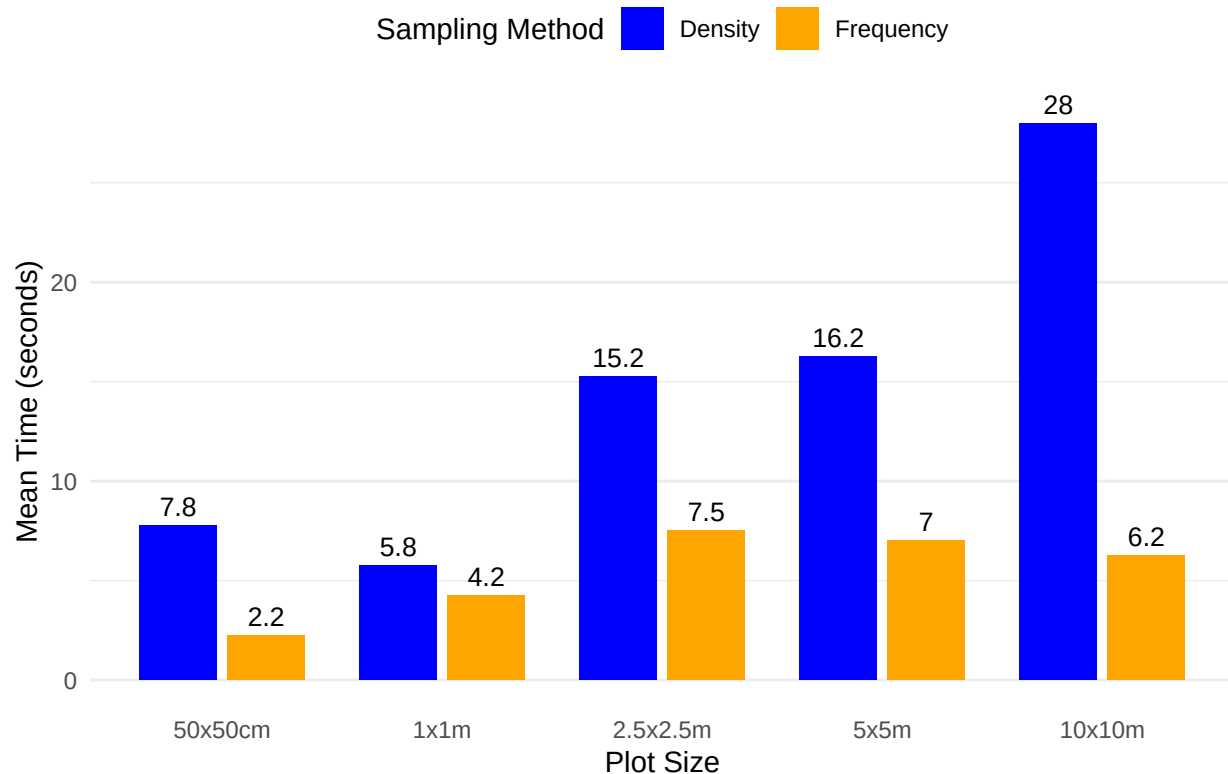
summary_time_frequency <- time_frequency_longer %>%
  group_by(plot_size) %>%
  summarise(mean_time = mean(count))

# 1. Combine both summaries and tag the sampling method
summary_combined <- bind_rows(
  summary_time_density %>% mutate(method = "Density"),
  summary_time_frequency %>% mutate(method = "Frequency")
)

# 2. Plot side-by-side bars using position_dodge()
ggplot(summary_combined, aes(x = plot_size, y = mean_time, fill = method)) +
  geom_col(position = position_dodge(0.8), width = 0.7) +
  geom_text(
    aes(label = round(mean_time, 1)),
    position = position_dodge(0.8),
    vjust = -0.5,
    size = 3.5
  ) +
  scale_fill_manual(
    name = "Sampling Method",
    values = c("Density" = "blue", "Frequency" = "orange")
  ) +
  labs(
    title = "Mean Sampling Time Comparison Across Plot Sizes",
    x = "Plot Size",
    y = "Mean Time (seconds)"
  ) +
  theme_minimal() +
  theme(
    panel.grid.major.x = element_blank(),
    legend.position = "top"
  )

```

Mean Sampling Time Comparison Across Plot Sizes



3.) In at least a paragraph, answer which is the most precise (least variable) plot size and why.

Standard error measures the precision in an estimate. Standard deviation measures the variability within the sample size. According to the standard error calculations, the quadrat size with the lowest standard error is the 50x50cm plot ($SE = 0.00$), followed by the 10x10m plot ($SE = 0.006$). The smallest quadrats have zero standard error because they all contained nothing. The largest plot having lower SE than the rest of the quadrats suggests that the population is not highly variable, and also follows the general rule that standard error decreases as sample size increases. Higher standard error means less precision. So, the most precise quadrat size is the 10x10m.

4.) In at least a paragraph, answer which is the most efficient plot size and why.

The most efficient plot size was the 50x50cm frequency plots. Frequency is easier and faster to measure since we aren't counting the number of individuals, only the presence or absence of them. The time spent trends upwards with increasing size of quadrat. The density quadrats take anywhere from twice as long to four times as long than the frequency quadrats.

- 5.) Converting the density to frequency, what plot size worked well for frequency?
- 6.) Theoretically frequency (p in %) and density (d) are related. Some suggest that the conversion from frequency to density is $d = -\ln(1-p/100)$. Calculate the estimated density for the 5x5m quadrat frequency data. (Convert to plants per m^2) Does the answer come out the same or different from the actual density answer?
- 7.) Write at least a paragraph commenting about this exercise.
- 8.) For the 2.5x2.5m plot size, calculate the sample size to produce, with 95% probability, a sample mean within 10% of the true population mean. What does this say about our ability to detect changes with the sample size of 20?